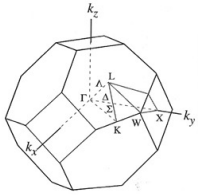
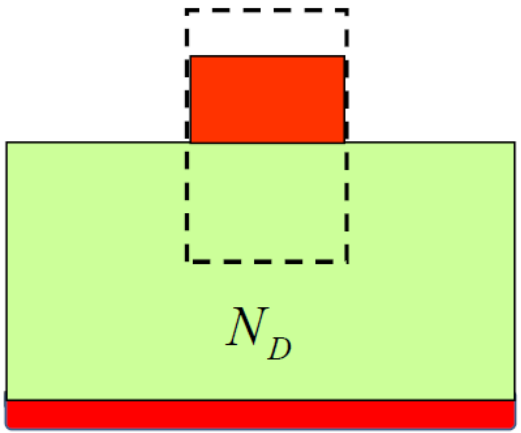
Schottky Diodes

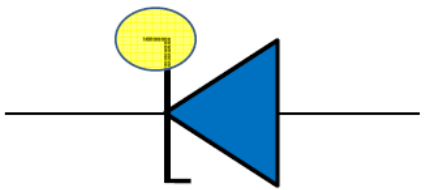


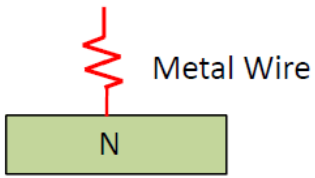
Metal Semiconductor Diode

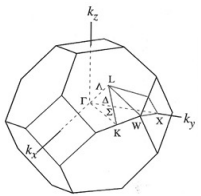


Symbols

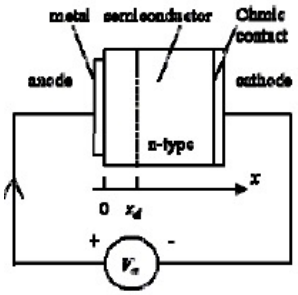








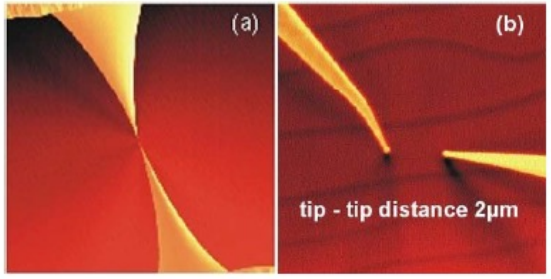
Applications of M-S Diode



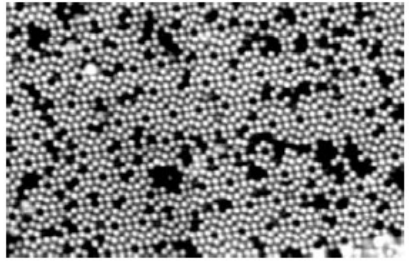
Detectors



STM on semiconductor



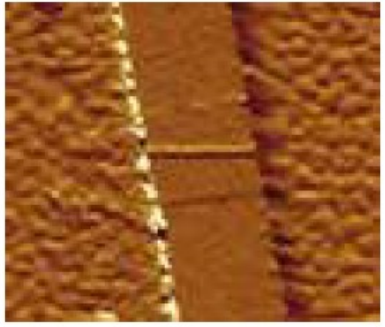
www.fz-juelich.de/ibn/index.php?index=674

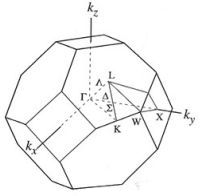


Original Bipolar

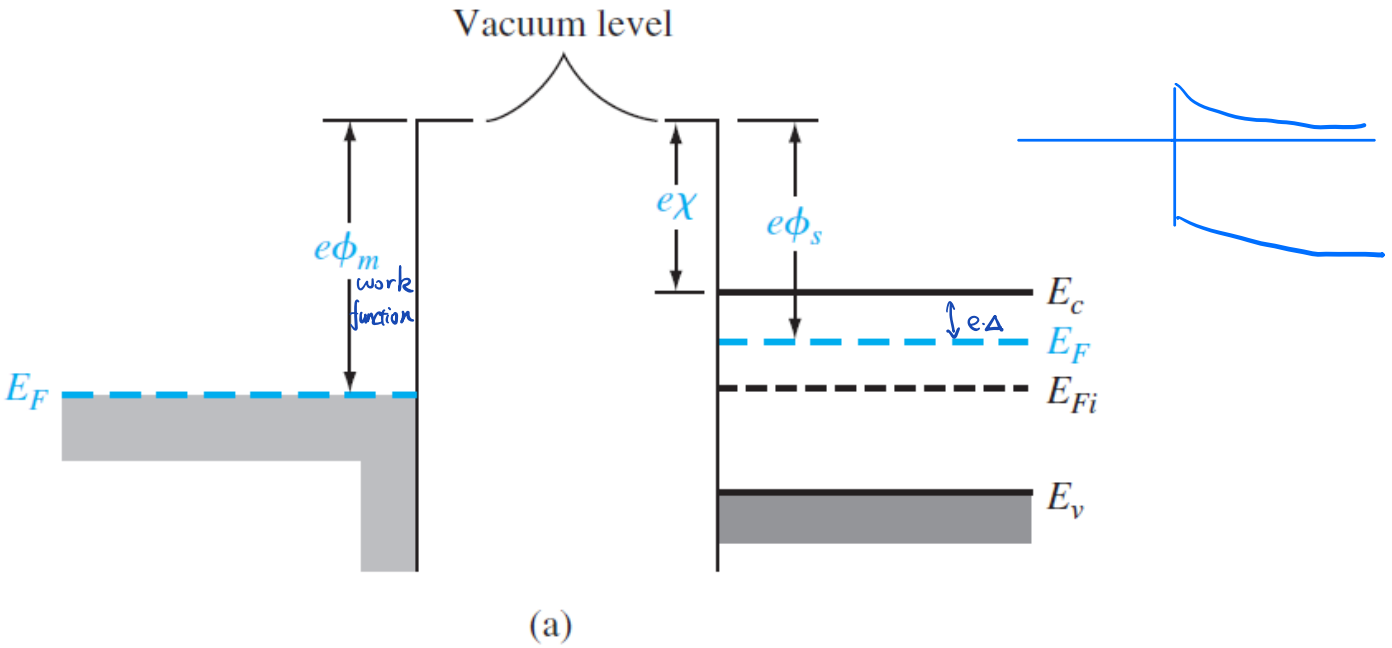


CNT Transistors

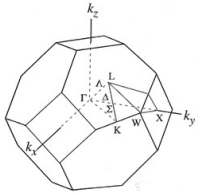


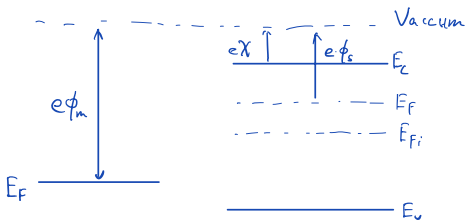


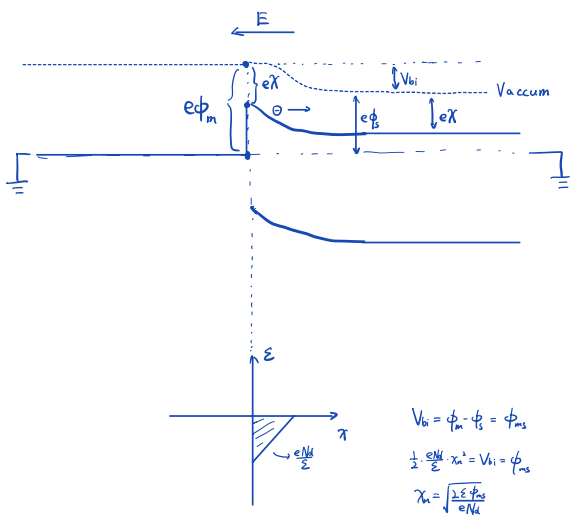
Band Diagram (before contact)



(a)







$V_{bi} = \phi_m - \phi_s = \phi_{ms}$

$\frac{1}{2} \cdot \frac{eN_d}{\varepsilon} \cdot x_n^2 = V_{bi} = \phi_{ms}$

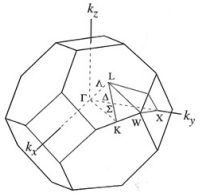
$x_n = \sqrt{\frac{2\varepsilon\phi_{ms}}{eN_d}}$

With a Bias: $V_{bi} \rightarrow V_{bi} - V$

Work Function of Some Metals

Table 9.1 | Work functions of some elements

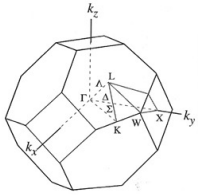
Element	Work function, ϕ_m
Ag, silver	4.26
Al, aluminum	4.28
Au, gold	5.1
Cr, chromium	4.5
Mo, molybdenum	4.6
Ni, nickel	5.15
Pd, palladium	5.12
Pt, platinum	5.65
Ti, titanium	4.33
W, tungsten	4.55



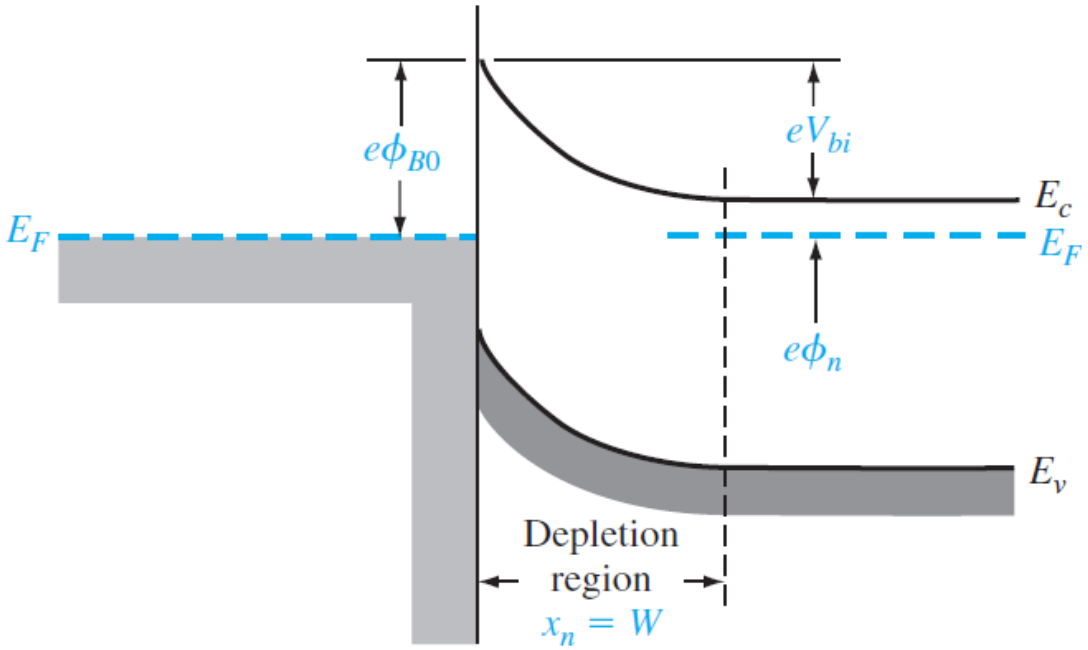
Electron Affinity

Table 9.2 | Electron affinity of some semiconductors

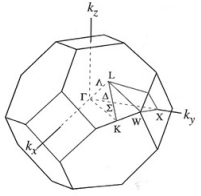
Element	Electron affinity, χ
Ge, germanium	4.13
Si, silicon	4.01
GaAs, gallium arsenide	4.07
AlAs, aluminum arsenide	3.5



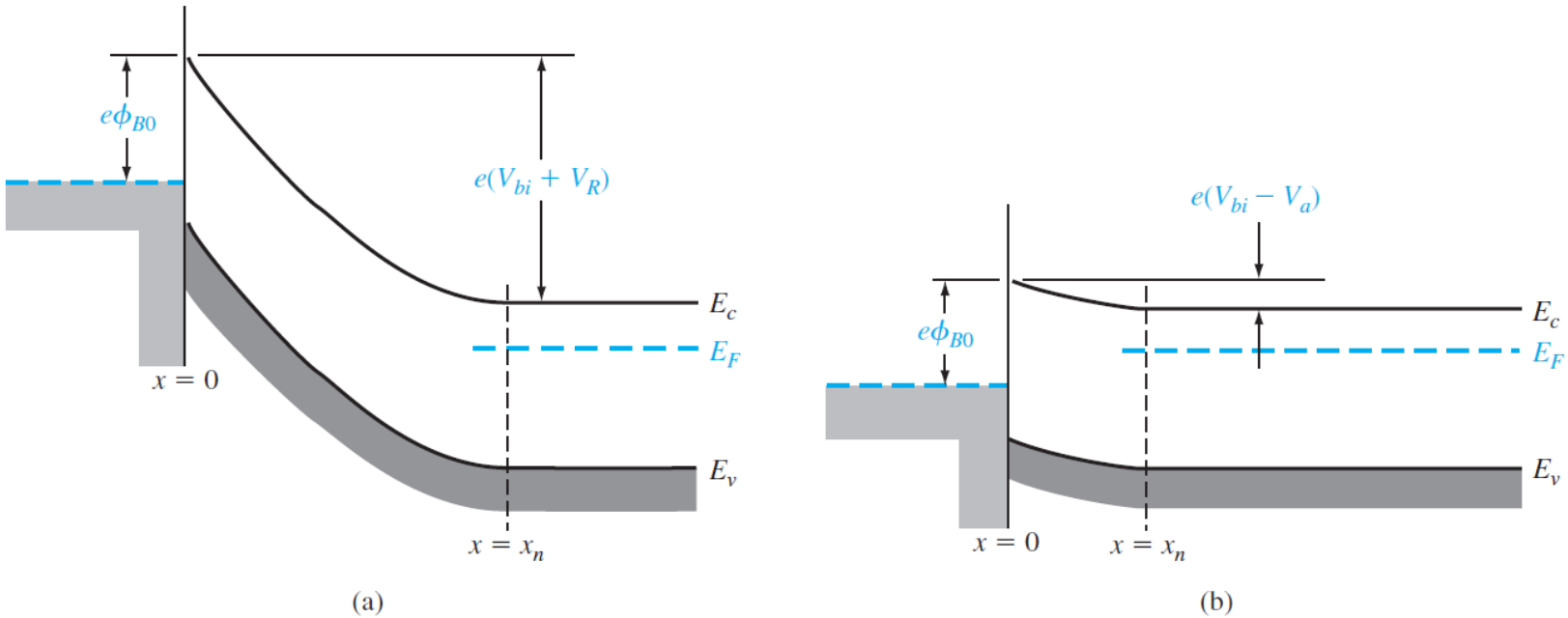
Band Diagram (after contact)



(b)



Band-diagram with Bias



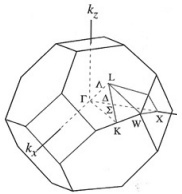
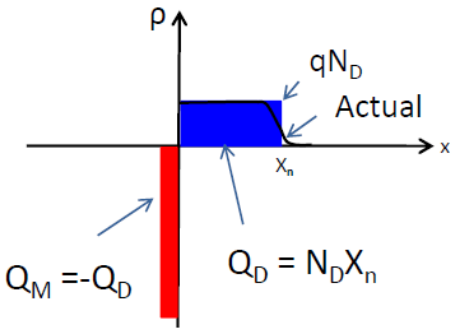
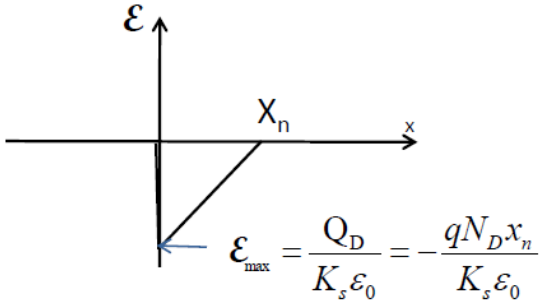


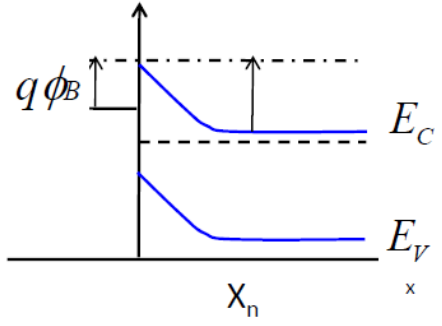
Figure 9.2 | Ideal energy-band diagram of a metal–semiconductor junction (a) under reverse bias and (b) under forward bias.

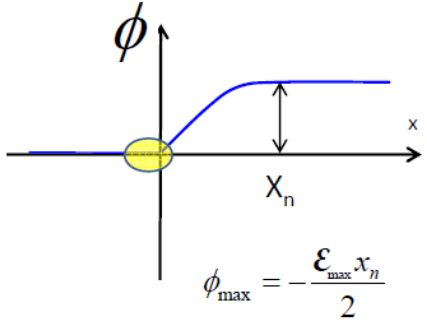
Complete Analytical Solution

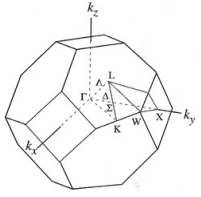
Depletion Approximation



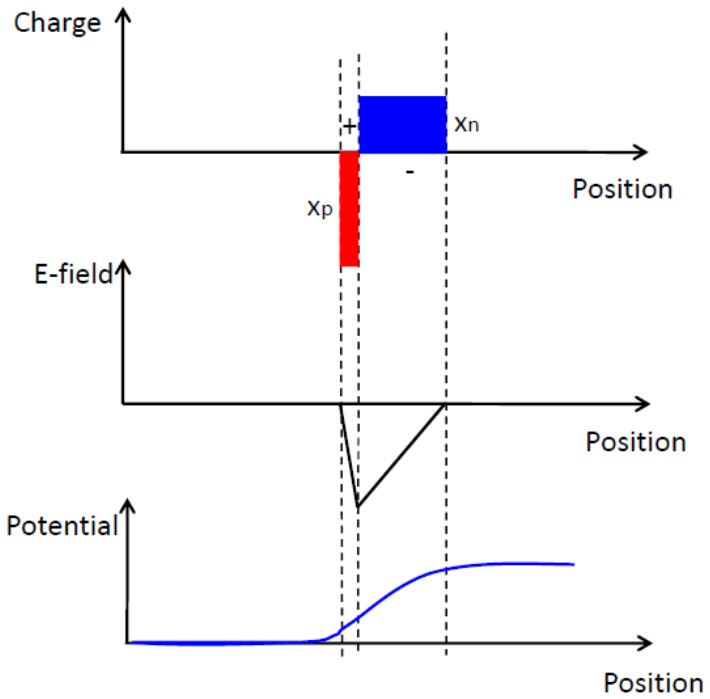








Analytical Solution



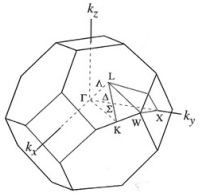
$$\mathcal{E}(0^+) = \frac{qN_D x_n}{k_s \varepsilon_0}$$

$$\mathcal{E}(0^-) = \frac{qN_M x_p}{k_s \varepsilon_0} \quad ?$$

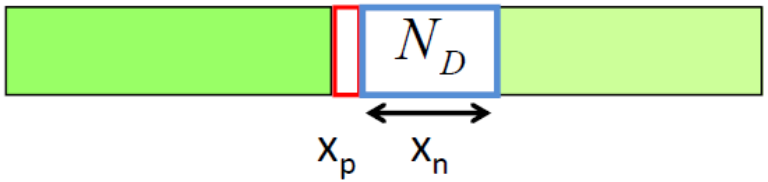
$$\Rightarrow N_D x_n = N_M x_p$$

$$qV_{bi} = \frac{\mathcal{E}(0^-)x_n}{2} + \frac{\mathcal{E}(0^+)x_p}{2}$$

$$= \frac{qN_D x_n^{\,2}}{2k_s \varepsilon_0} + \frac{qN_M x_p^{\,2}}{2k_s \varepsilon_0}$$



Depletion Region

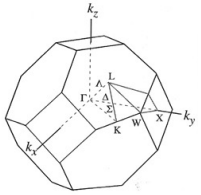


$$N_D x_n = N_M x_p$$

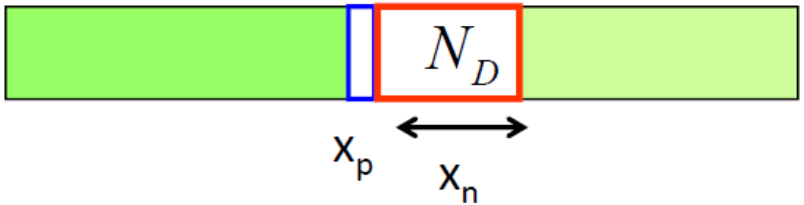
$$qV_{bi} = \frac{qN_D x_n^{\,2}}{2k_s\varepsilon_0} + \frac{qN_M x_p^{\,2}}{2k_s\varepsilon_0}$$

$$x_n = \sqrt{\frac{2k_s\varepsilon_0}{q}\frac{N_M}{N_D\left(N_M + N_D\right)}V_{bi}} \rightarrow \sqrt{\frac{2k_s\varepsilon_0}{q}\frac{1}{N_D}V_{bi}}$$

$$x_p = \sqrt{\frac{2k_s\varepsilon_0}{q}\frac{N_D}{N_M\left(N_M + N_D\right)}V_{bi}} \rightarrow 0$$

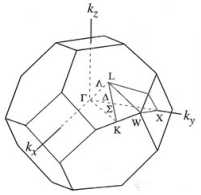


Depletion Regions with Bias

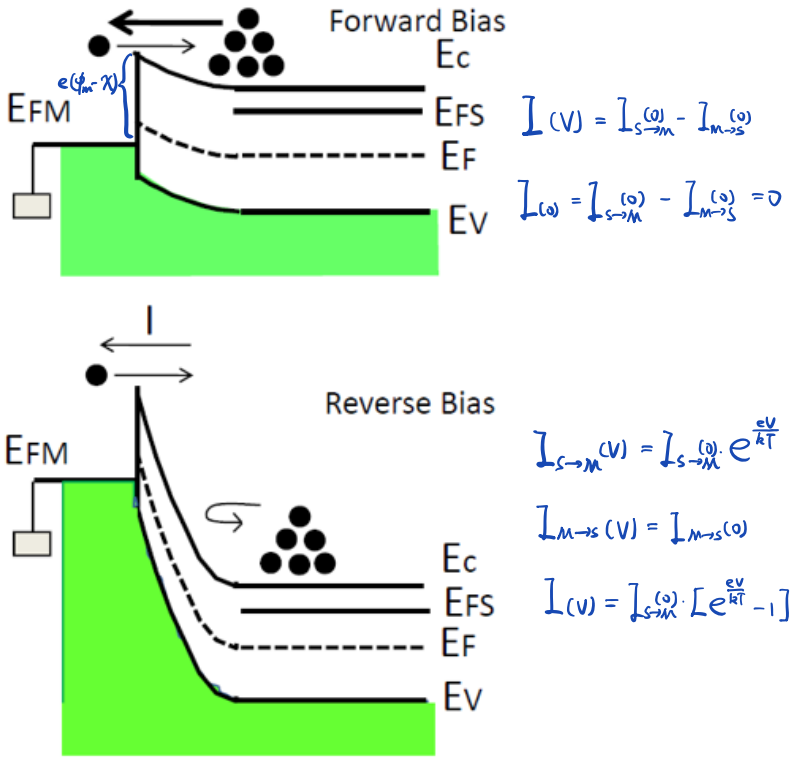


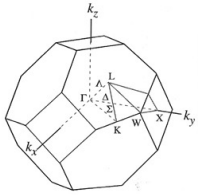
$$x_n = \sqrt{\frac{2k_s\varepsilon_0}{q}\frac{N_M}{N_D\left(N_M + N_D\right)}V_{bi}} \rightarrow \sqrt{\frac{2k_s\varepsilon_0}{q}\frac{1}{N_D}\left(V_{bi} - V_A\right)}$$

$$x_p = \sqrt{\frac{2k_s\varepsilon_0}{q}\frac{N_D}{N_M\left(N_M + N_D\right)}V_{bi}} \rightarrow 0$$

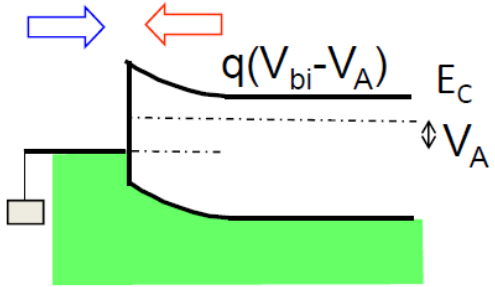


I-V Characteristics





Left Boundary Condition



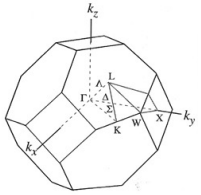
$$J_T(V_A) = J_{m\to s}(V_A) - J_{s\to m}(V_A)$$
$$= J_{m\to s}(0) - J_{s\to m}(V_A)$$

$$J_T(V_A = 0) = 0 = J_{m\to s}(0) - J_{s\to m}(0)$$

(detailed balance)

$$\Rightarrow J_{m\to s}(0) = J_{s\to m}(0)$$

$$J_T(V_A) = J_{s\to m}(0) - J_{s\to m}(V_A)$$

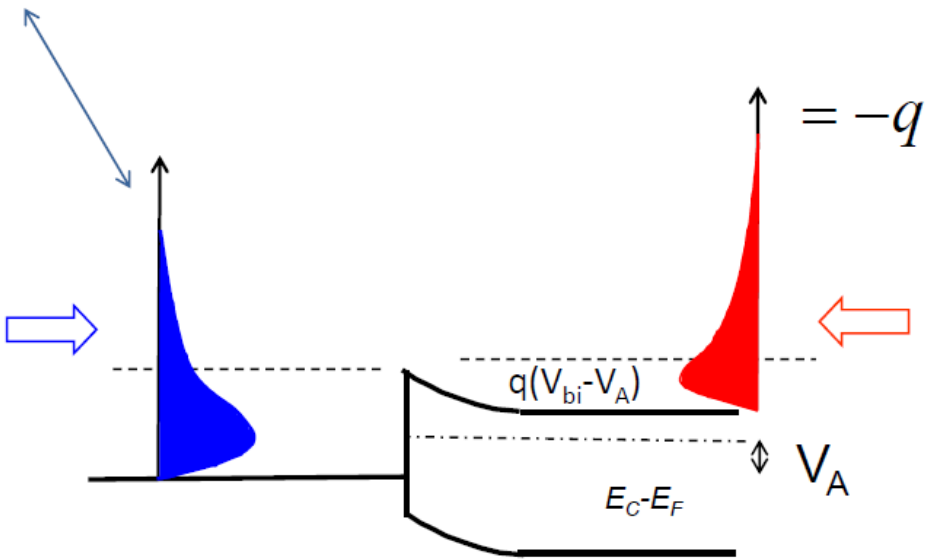


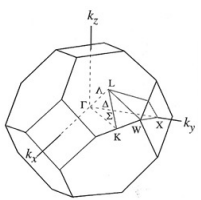
Semiconductor to Metal Flux

$$J_{m\to s}(V_A) = -q\frac{n_m}{2}e^{-\frac{q\Phi_B}{kT}}\upsilon_{th} \qquad J_{s\to m}(V_A) = -q\frac{n_s}{2}e^{-q\frac{V_{bi}-V_A}{kT}}\upsilon_{th}$$

$$= -q\frac{n_s\upsilon_{th}}{2}e^{-\frac{qV_{bi}}{kT}} \times e^{\frac{qV_A}{kT}}$$

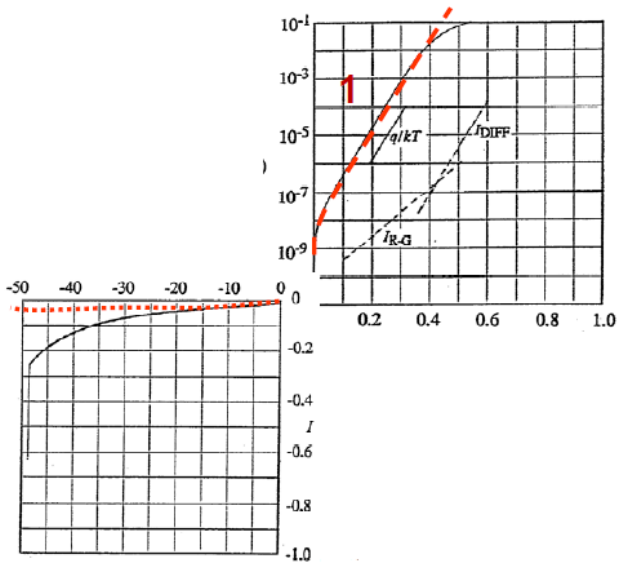
$$= -q\frac{n_m\upsilon_{th}}{2}e^{-\frac{q\Phi_B}{kT}}e^{\frac{qV_A}{kT}}$$

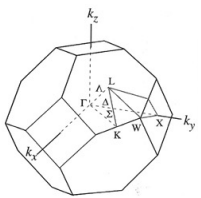




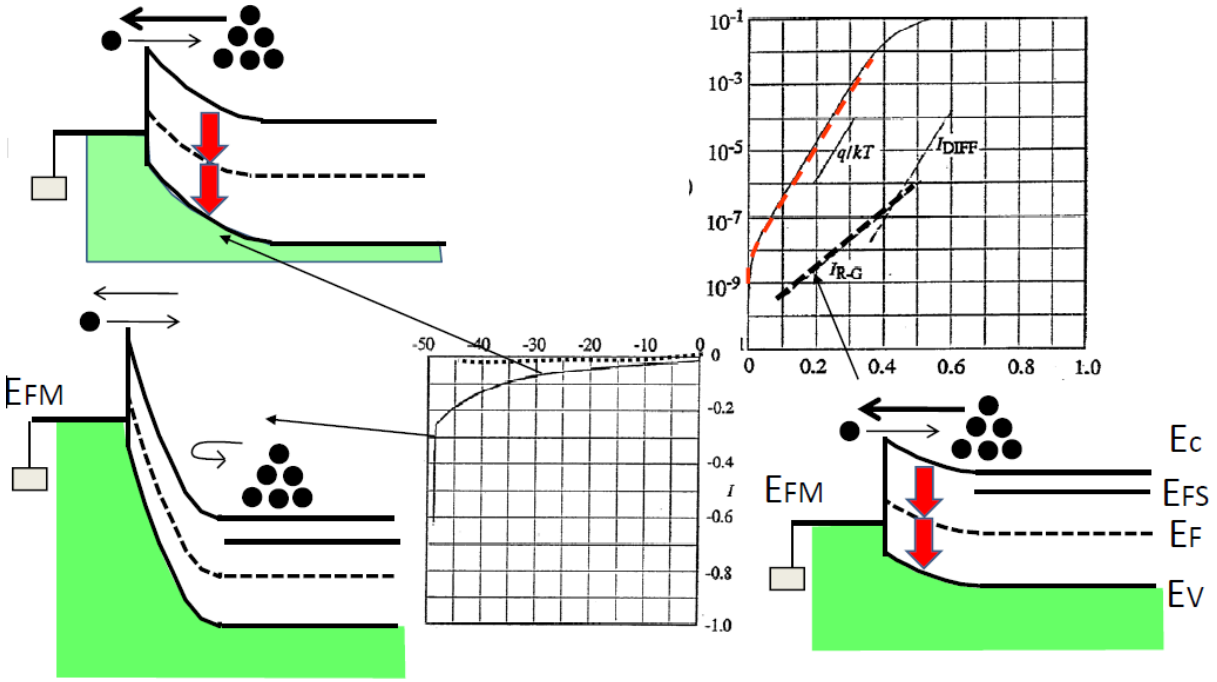
Total Flux

$$J_T = J_{s\to m}(0) - J_{s\to m}(V_A) = \frac{qn_m\upsilon_{th}}{2} e^{\frac{-q\Phi_m}{kT}} \left[e^{\frac{qV_A}{kT}} - 1 \right]$$

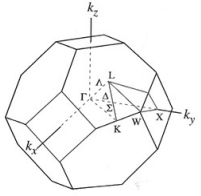




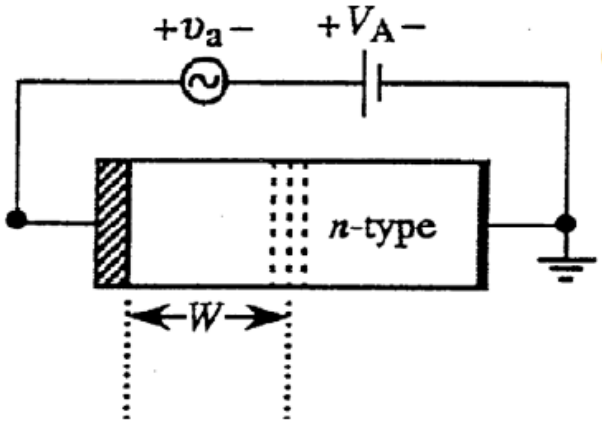
Recombination-Generation/Impact-ionization

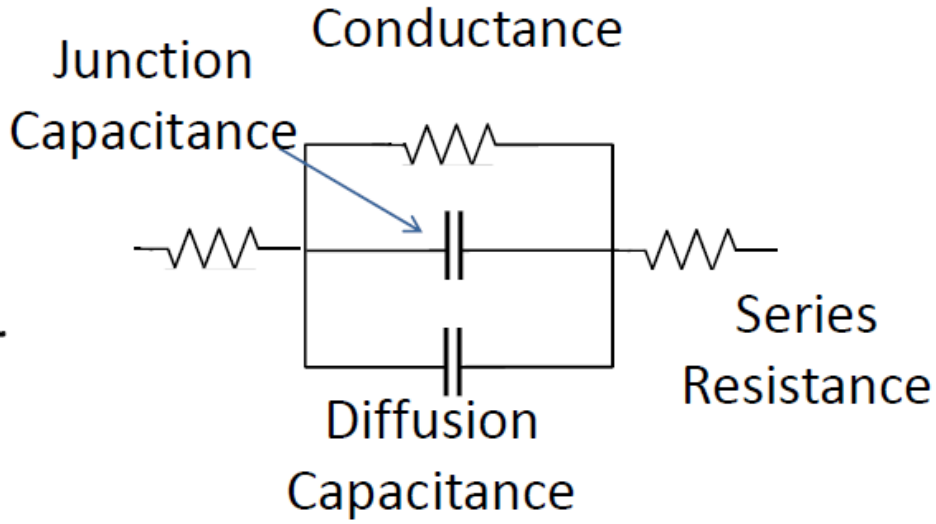


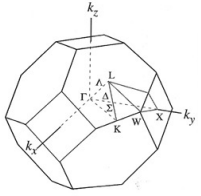
SAME technique as in p-n junction except integrate to x_n only



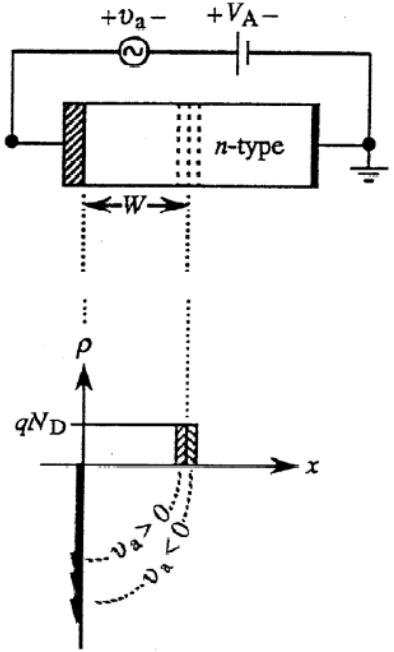
AC Response

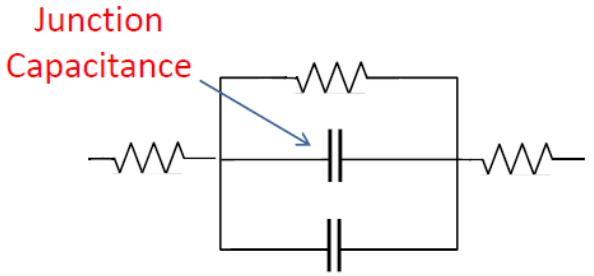

+ v_a −
+ V_A −
n-type
W


Conductance
Junction
Capacitance
Series
Resistance
Diffusion
Capacitance



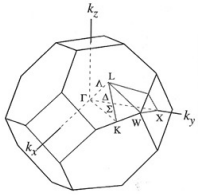
Junction Capacitance (Majority Carriers)



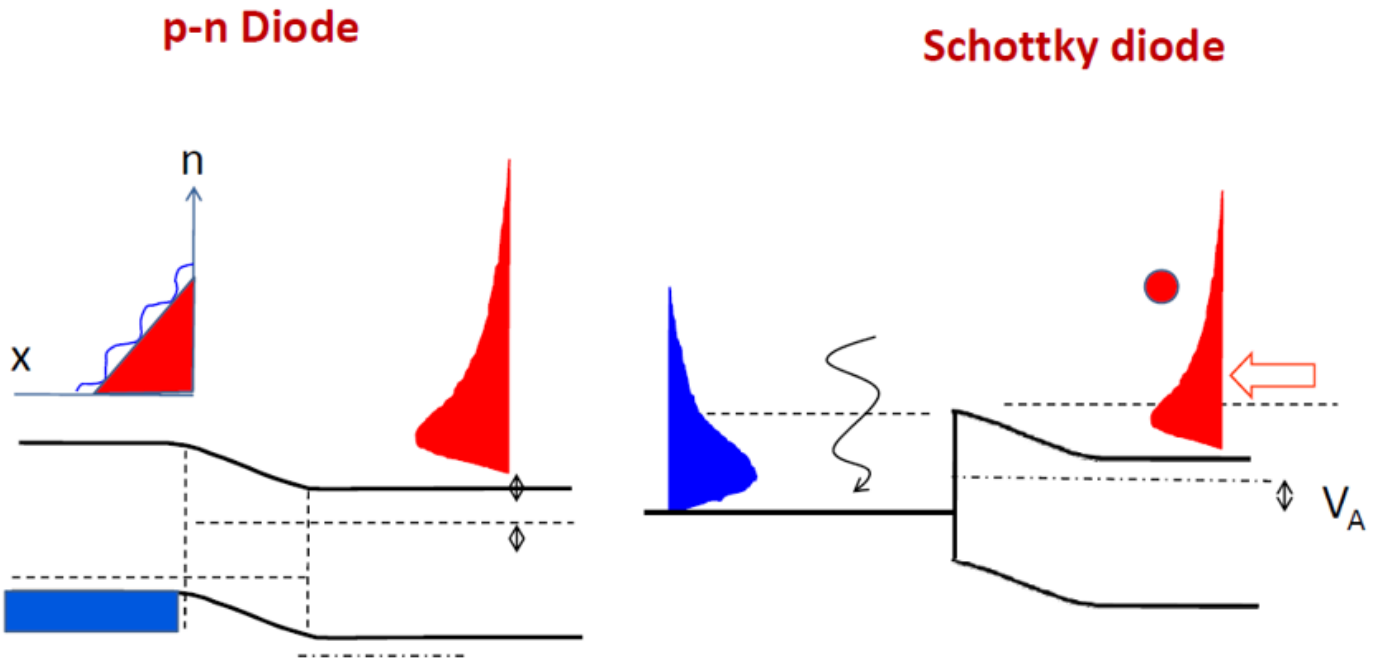


$$C_J = \frac{\kappa_s \varepsilon_0 A}{W}$$

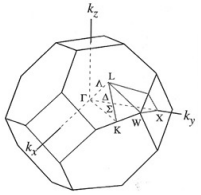
$$C_J = \frac{\kappa_s \varepsilon_0 A}{\sqrt{\frac{2\kappa_s \varepsilon_0}{qN_D}(V_{bi} - V_A)}}$$



No Diffusion Capacitance in Schottky Diodes

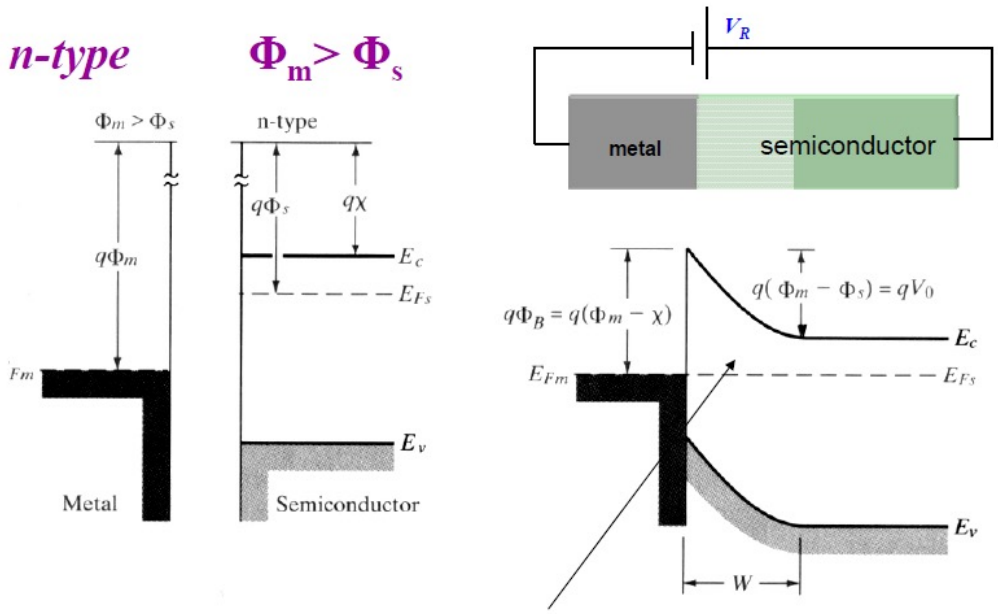


No minority carrier transport and therefore no diffusion capcitance

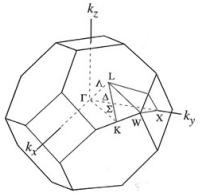


Schottky contacts

n-type semiconductor



Rectifying Schottky contact creates an electron depletion region at the metal-semiconductor interface

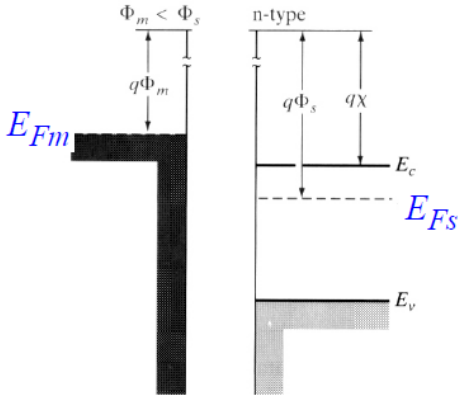


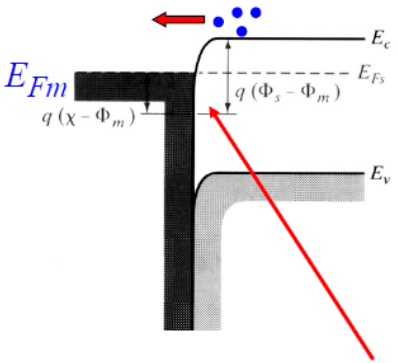
Ohmic Contact

Ohmic Contact to n-type semiconductor

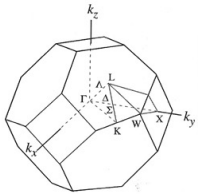
$\Phi_m < \Phi_s$

Majority carriers are electrons; there is no potential barrier for electrons in both forward or reverse directions:

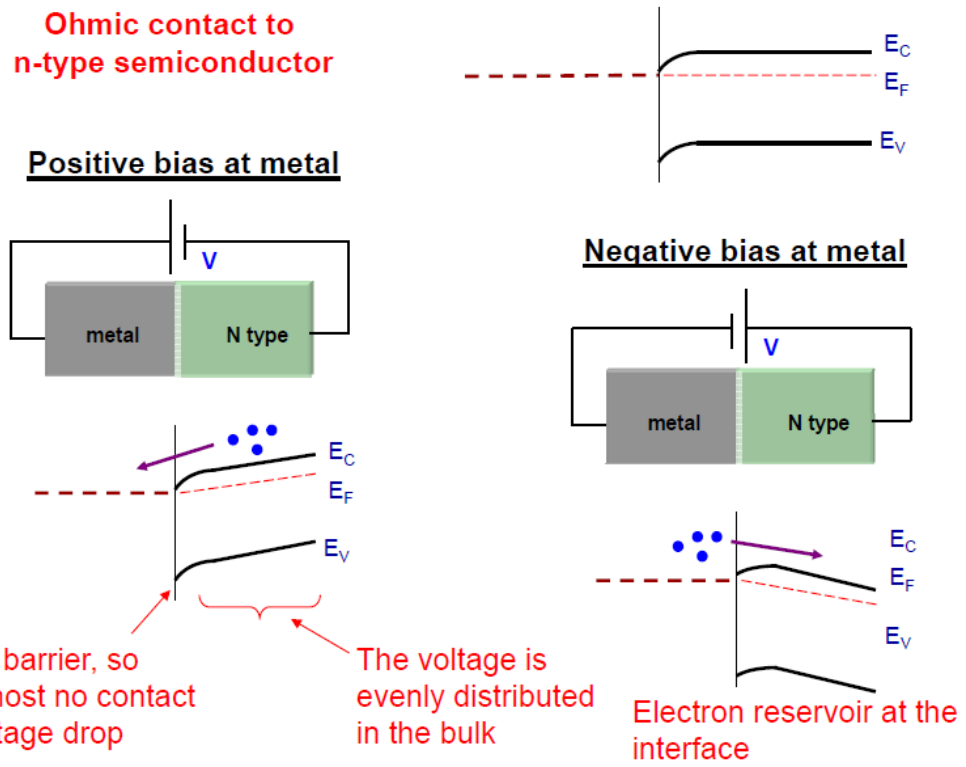


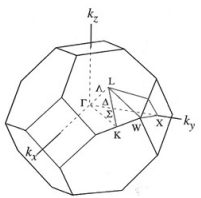


Non-rectifying Schottky contact creates an electron ***accumulation*** region at the metal-semiconductor interface.
The electron concentration in the contact region is ***higher*** than that in the bulk. The resistance of the contact region is ***low***.

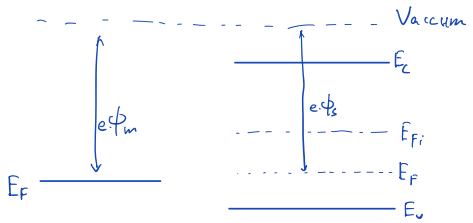


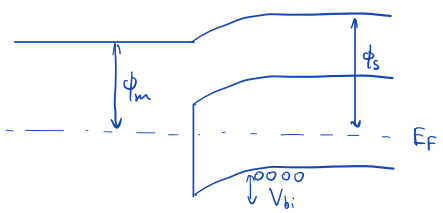
Ohmic Contact Under Bias





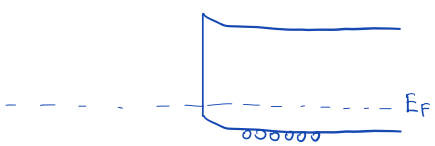
M - P





$V_{bi} = \phi_m - \phi_s = \phi_{ms}$

$\phi_s > \phi_m$: schottky contact.



$\phi_s < \phi_m$: ohimic contact

Schottky Contacts v.s Ohmic Contacts

Schottky contacts
(Rectifying contacts)

Criteria:
- n-type

$\Phi_m > \Phi_s$
- p-type

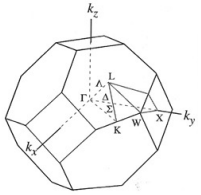
$\Phi_m < \Phi_s$

Ohmic Contacts
(Non-rectifying contacts)

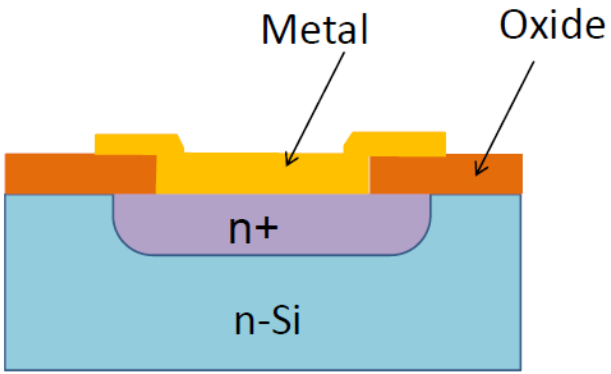
Criteria:
- n-type

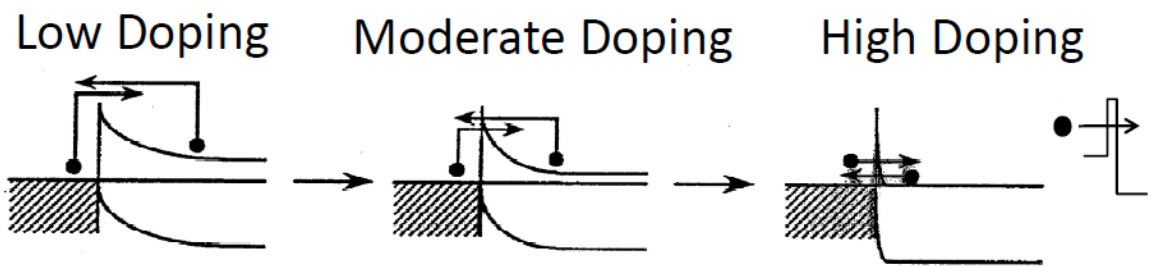
$\Phi_m < \Phi_s$
- p-type

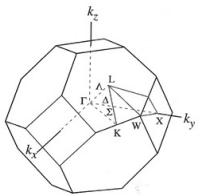
$\Phi_m > \Phi_s$



Ohmic Contact vs. Schottky Contacts







Conclusion

- Schottky diodes have wide range of applications in practical devices
- The key distinguishing feature of Schottky diode is that it is a majority carrier device
- Schottky Contacts v.s Ohmic Contacts

